

Halvor Hafnor

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SUMMARY

Embedded software and machine learning engineer building classification pipelines for real sensor data on constrained hardware, from collection through feature extraction to on-device inference. Two years at Aerendir Mobile, a shipped automotive product, a pending patent, and an M.S. at UC San Diego.

EXPERIENCE

Embedded Software and ML Engineer

Jul. 2024 – Present

Aerendir Mobile · part time in term, full time out of term

Mountain View, CA

- Improved the robustness of the anti-bot liveness gate behind **A-Live**, building a lab that evaluates the motion-sensor classifiers against adversarial input and quantifies what each added defense is worth.
- Built Android applications for real-time biometric data streaming and workflow control.
- Collected and classified the motion dataset behind the company's biometric authentication work: 400 participants, each recorded holding the phone in three postures, sitting, standing and walking, from the accelerometer and gyroscope.
- Prototyped automotive driver monitoring from steering-wheel biometrics, shipped as **AerWheel**. Trained decision-tree classifiers in scikit-learn on features from a 240 Hz six-axis IMU, first through ST's on-sensor Machine Learning Core and then as an external tree running in firmware. Better than **99% accuracy** at under 0.5% false positives, answering within 1.5 seconds. Subject of a pending patent naming me as an inventor.

Electrical Engineering Intern

Jun. 2021 – Sep. 2023

ABB Electrification

Skien, Norway

- Assembled and wired around 300 industrial control panels and power cabinets from engineering schematics, across three consecutive summers.
- Supported production and testing of custom electrical systems for industrial clients.

EDUCATION

University of California - San Diego - San Diego, CA

Computer Engineering, M.S.

Jan. 2026 – June 2027

Computer Engineering, B.S. Major GPA: 3.9

Sep. 2024 – June 2026

- **Provost Honors** every quarter enrolled, and **Tau Beta Pi**, the national engineering honor society.

De Anza College - Cupertino, CA

GPA: 4.0

Computer Science & Engineering

Sep. 2022 – June 2024

- Math & Science Tutor and CS Teaching Assistant for calculus, physics, discrete math, C++, Java and Assembly.

TECHNICAL SKILLS

Languages: C++, C, Python, TypeScript, Java, Bash, ARM Assembly

Embedded: Embedded Linux, Android, QML/Qt, STM32CubeIDE, ST sensor datalogging, on-sensor inference, Docker

Signals & ML: PyTorch, scikit-learn, NumPy, pandas, SciPy, OpenCV, YOLOv8, signal processing, feature extraction

Systems: Node, SQLite, WebSocket protocols

Agentic Development: Claude Code, MCP, spec-driven work breakdown, subagent orchestration, verification gates

PROJECTS

TabTerm, a macOS terminal in Chrome tabs

- A Chrome extension over a background terminal daemon that puts every shell session in a real Chrome tab, so tab groups, window management and keyboard shortcuts apply to terminals. Adds a command launcher, history search, project-scoped saved commands and a plugin API.

Laser targeting on an autonomous car, UC San Diego, ECE 148

- Real-time person tracking on a Raspberry Pi 5, in Docker. YOLOv8 detection and OAK-D stereo depth, converted through a 71.9° field of view into pan and tilt angles so two servos hold a laser on the nearest person from 0.5 to 2.9 m. Depth is the median of a 10x10 patch, since one bad pixel throws the aim.